COMPLETE ELEVEN-VERTEX MINIMAL EXAMPLES SEPARATING COHEN-MACAULAYNESS, SHELLABILITY AND VERTEX DECOMPOSABILITY

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Editorial note. Complete original author mathematical paper, native diagrams/tables and embedded bibliography retained unchanged. Authors retain responsibility for mathematics; no independent review, proof certification, counterexample search, added hypothesis or proof repair.

Editorial note. Original exhaustive census is author evidence, with full original method/counts and proofs retained; it has not been rerun or independently certified. The author reports both a 2022–23 Macaulay2 run and an AI-assisted-code 2026 cross-check, with identical examples. No agent computation or validation claim is made.

Editorial note. Original AI declaration721–724 and computational disclosure666/697–704 retained: figures/tables/writing and code cross-check assistance. This is not treated as explicit discovery of the selected core proof by AI; no AI-free claim or author-process audit.

Editorial note. Only the complete minimal-example classification counts. Individual eight graphs, characteristic cases, expansion/inflation theorem and larger-n consequences are not split into additional counts.

Editorial note. Standalone wrapper boundary excluded accurately: original line1105 is end-document, while original mathematics and embedded bibliography close by line1104. Raw source unchanged; no mathematical or proof material omitted.

Allowed standard prerequisites (editorial).

- Well-covered graph structures and independence complexes. All original definitions, exact graph-class/connectedness/isolated-vertex and field-characteristic hypotheses, original supporting statements and citations are retained with complete source-local arguments.. Full author paper context and native bibliography supplied; no proof reconstruction, assumption repair or independent certification.

1. Introduction

One theme in the study of combinatorial commutative algebra is to determine when the independence complex $\operatorname{Ind}(G)$ of a graph G (sometimes called a flag complex) has additional topological or combinatorial structure, such as being *Cohen-Macaulay*, *shellable*, or *vertex decomposable* (see Section 2 for definitions). Indeed, Villarreal's paper [26], one of the founding papers in the field, described conditions on G that force $\operatorname{Ind}(G)$ to be Cohen-Macaulay. For these three properties, the following implications always hold:

$$(\star) \qquad \operatorname{Ind}(G) \text{ is vertex decomposable} \Rightarrow \operatorname{Ind}(G) \text{ is shellable} \Rightarrow \operatorname{Ind}(G) \text{ is Cohen-Macaulay.}$$

In fact, these implications hold for any simplicial complex Δ, not just independence complexes. While being vertex decomposable and shellable are combinatorial properties, the Cohen-Macaulay property is defined in terms of reduced simplicial homology over a field $\mathbb{K}$, or equivalently, the length of the longest regular sequence in the associated Stanley-Reisner ring. Thus Cohen-Macaulayness also depends upon the characteristic of $\mathbb{K}$. If one adds additional hypotheses on G, it can be shown that the converse implications of ($\star$) also hold, e.g., if G is a bipartite graph [25], a chordal graph [7, 11, 28], a well-covered graph with girth ≥ 5 [3], or a very well-covered graph [14], among other families.

In general, the reverse implications of ($\star$) fail. Moriyama–Takeuchi [17] gave minimal examples of simplicial complexes on six vertices that are shellable, but not vertex decomposable. However, these complexes are not the independence complex of any graph; to the best of our knowledge, the first published example of a graph G where $\operatorname{Ind}(G)$ is shellable but not vertex decomposable is a circulant

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graph on 16 vertices found by Earl–Vander Meulen–Van Tuyl [8]. Using a construction of Moradi–Khosh-Ahang [16], Vander Meulen–Van Tuyl [24] leveraged this circulant graph to show that for all $n \geq 16$, there is a graph on n vertices where the first reverse implication fails. On the other hand, Baker–Vander Meulen–Van Tuyl [2] showed via a computer search that for all graphs on $n \leq 10$ vertices, the implications of ($\star$) can be reversed.

This leaves the tantalizing question of what are the minimal graphs for which ($\star$) fails, which must necessarily have at least 11 vertices. We report on an exhaustive computer search to fill this gap. The following theorem summarizes the findings of our computer search.

Theorem 1.1. *We have the following classification:*

(1) The six graphs $H_1, H_2, \ldots, H_6$ defined in Section 3 are the only graphs on 11 vertices with $\operatorname{Ind}(G)$ *shellable but not vertex decomposable.*
(2) If $\operatorname{char}(\mathbb{K}) \neq 2$*, then the two graphs G_1 and G_2 defined in Section 3 are the only graphs on 11 vertices with* $\operatorname{Ind}(G)$ *Cohen-Macaulay but not shellable.*
(3) If $\operatorname{char}(\mathbb{K}) = 2$*, then for all graphs on 11 vertices,* $\operatorname{Ind}(G)$ *is Cohen-Macaulay if and only if* $\operatorname{Ind}(G)$ *is shellable.*

We have organized this short note as follows. Section 2 recalls the relevant definitions, Section 3 provides the main examples plus some additional commentary on our examples, and Section 4 summarizes our computer search.

2. Background

A *simplicial complex* Δ on a set of vertices $X = \{1, \ldots, n\}$ is a subset of the power set of X such that: (1) if $F \in \Delta$ and $H \subseteq F$, then $H \in \Delta$; and (2) $\{i\} \in \Delta$ for all $i \in X$. Note that $\emptyset \in \Delta$. An element $F \in \Delta$ is called a *face* and the maximal faces are called the *facets*. If $F_1, \ldots, F_s$ is a complete list of facets of Δ, then we write $\Delta = \langle F_1, \ldots, F_s \rangle$.

The *dimension* of a face $F \in \Delta$ is given by $\dim(F) = |F| - 1$; we then define the *dimension of* Δ to be $\dim(\Delta) = \max\{\dim(F) \mid F \in \Delta\}$. A simplicial complex Δ is *pure* if $\dim(F) = \dim(\Delta)$ for all of its facets. If $\dim(\Delta) = d$, then the *f-vector* of Δ is the $(d+2)$-tuple $f(\Delta) = (f_{-1}, f_0, \ldots, f_d)$ where $f_i = |\{F \in \Delta \mid \dim(F) = i\}|$. We use the convention that $\dim(\emptyset) = -1$, and thus $f_{-1} = 1$. For $\ell = 0, \ldots, d+1$, set $h_\ell = \sum_{i=0}^{\ell} (-1)^{\ell - i} \binom{d+1-i}{\ell - i} f_{i-1}$. The tuple $h(\Delta) = (h_0, \ldots, h_{d+1})$ is the *h-vector* of Δ.

For a simplicial complex Δ on X and a fixed $i \in X$, the *link* of $\{i\}$ in Δ is the subcomplex $\operatorname{link}_\Delta(i) = \{F \in \Delta \mid F \cup \{i\} \in \Delta,\ F \cap \{i\} = \emptyset\}$, and the *deletion* is the simplicial complex $\operatorname{del}_\Delta(i) = \{F \setminus \{i\} \mid F \in \Delta\}$. The deletion and link are simplicial complexes on $X \setminus \{i\}$.

For a fixed field $\mathbb{K}$, the *Stanley-Reisner ideal* of Δ on $X = \{1, \ldots, n\}$ is the monomial ideal

$$I_\Delta = \langle x_{i_1} \cdots x_{i_s} \mid \{i_1, \ldots, i_s\} \notin \Delta \rangle \subseteq R = \mathbb{K}[x_1, \ldots, x_n].$$

The quotient ring R/I_Δ is the *Stanley-Reisner ring* of Δ. It is known (e.g. [27, Corollary 6.3.5]) that $K\text{-}\dim(R/I_\Delta) = \dim(\Delta) + 1$, where $K\text{-}\dim(R/I_\Delta)$ denotes the Krull dimension of R/I_Δ.

We now define the special families of simplicial complexes studied in this paper.

Definition 2.1. Let Δ be a pure simplicial complex on X with $\dim(\Delta) = d$.

(1) Then Δ is *vertex decomposable* if either: (a) $\Delta = \langle F \rangle$ has a single facet, equivalently, Δ is a *simplex*, or (b) there exists a vertex $i \in X$ such that $\operatorname{link}_\Delta(i)$ and $\operatorname{del}_\Delta(i)$ are vertex decomposable.
(2) Then Δ is *shellable* if there exists an ordering $F_1 \prec F_2 \prec \cdots \prec F_s$ of the facets of Δ such that for all $1 \leq j < i \leq s$, there exists some $a \in F_i \setminus F_j$ and some $k \in \{1, \ldots, i-1\}$ such that $\{a\} = F_i \setminus F_k$.
(3) Then Δ is *Cohen-Macaulay* if the longest regular sequence in R/I_Δ has length $d+1$, that is, there exist $f_1, \ldots, f_{d+1} \in R$ such that $\overline{f}_1$ is a non-zero divisor of R/I_Δ, and for $j = 2, \ldots, d+1$, the element $\overline{f}_j$ is a non-zero divisor of $R/(I_\Delta + \langle f_1, \ldots, f_{j-1} \rangle)$, where $\overline{f}_k$ denotes the appropriate equivalence class.

Remark 2.2. Vertex decomposability was introduced by Provan–Billera [23]. The length of the longest regular sequence of a ring is its *depth*, so the Cohen-Macaulay definition is the requirement that the depth of R/I_Δ equal its Krull dimension. An equivalent definition for Cohen-Macaulay simplicial complexes is to use Reisner's criterion in terms of the reduced simplicial homology of Δ over a field $\mathbb{K}$ (see [27, Theorem

5.3.5]). While vertex decomposability and shellability only depend on the combinatorics of Δ, the Cohen-Macaulay property of Δ also depends upon the characteristic of $\mathbb{K}$.

We have the following properties, which also justify the implications ($\star$) in the introduction.

Theorem 2.3. *Let Δ be a pure simplicial complex on X with $\dim(\Delta) = d$.*

(1) [23, Corollary 2.9] *If Δ is vertex decomposable, then Δ is shellable.*
(2) [27, Theorem 5.3.18] *If Δ is shellable, then Δ is Cohen-Macaulay (over any field $\mathbb{K}$).*
(3) [27, Theorem 5.4.8] *If Δ is Cohen-Macaulay, then $h(\Delta)$ has no negative entries.*

Moving forward, we focus primarily on the independence complex of a graph. Given a finite simple graph $G = (V, E)$ with vertex set $V = \{1, \ldots, n\}$ and edge set E, we say $W \subseteq V$ is an *independent set* if for every edge $e \in E$, $e \not\subseteq W$. The complement of an independent set is a *vertex cover*. The *independence complex* of G is

$$\mathrm{Ind}(G) = \{W \subseteq V \mid W \text{ is an independent set}\}.$$

Then $\mathrm{Ind}(G)$ is a simplicial complex on the vertex set $V = \{1, \ldots, n\}$. The complex $\mathrm{Ind}(G)$ is pure if and only if the graph G is *well-covered*, that is, every minimal vertex cover has the same size. The Stanley-Reisner ideal of $\mathrm{Ind}(G)$ is the *edge ideal* of G, that is, $I(G) = \langle x_i x_j \mid \{i, j\} \in E\rangle$ in $R = \mathbb{K}[x_1, \ldots, x_n]$.

3. The Examples

Table 1 lists the eight connected graphs $G_1, G_2, H_1, \ldots, H_6$ on 11 vertices found in our exhaustive search. These eight graphs are the minimal examples listed in Theorem 1.1 that demonstrate that the reverse implications of ($\star$) fail to hold in general. Graphs G_1 and G_2 are drawn in Figure 1 and graphs $H_1, \ldots, H_6$ all appear in Figure 2. We have also included the graph6 representation of each graph below Table 1.

	CM, not shellable		Shellable, not VD					
v	G_1	G_2	H_1	H_2	H_3	H_4	H_5	H_6
1	4,5,7,8	4,5,8,9	4,6,7,9	4,6,7,9	4,6,8,9	4,6,8,9	4,6,8,9	4,6,9,10
2	5,6,8,10,11	5,6,8,10,11	5,8,9,10	5,8,9,10	5,7,9,10,11	5,7,9,10	5,7,9,10,11	5,7,8,9
3	6,7,9,10	6,7,9,10	6,7,9,10,11	6,7,10,11	6,7,10,11	6,7,10,11	6,7,10,11	6,7,8,11
4	8,9,10,11	7,8,11	7,8,10,11	7,8,9	7,8,9	7,8,9	7,8,9	8,9,10,11
5	7,9,11	9,10,11	8,10,11	8,10,11	8,9,10	8,9,10,11	8,9,10	7,9,10,11
6	8,9	8,9,11	9,11	9,10,11	10,11	10,11	10,11	9,10,11
7	10	10,11	10	10,11	8,11	8,11	8,11	10
8	–	–	11	11	11	11	–	11
9	11	11	–	–	–	–	11	–
10	11	–	–	–	–	–	–	–
$\lvert E\rvert$	25	25	24	24	24	24	24	25

graph6 codes

G_1 `JCpdUg{[ap_` G_2 `JCp`eikYa|?` $H_1$ `JCQdarc]`y?` H_2 `JCQdarSZ`]?`
H_3 `JCQbcvWZBM?` H_4 `JCQbcvWZ@]?` H_5 `JCQbcvWZBL?` H_6 `JCQbRb[f`y?`

TABLE 1. The eight graphs of Theorem 1.1. The entry in row v lists the neighbours $w > v$ of vertex v, so each edge $\{v, w\}$ with $v < w$ appears exactly once. Below the table, each graph is also given in graph6 format, with vertex i of the table corresponding to vertex $i - 1$ in the graph6 encoding.

Example 3.1 (Cohen-Macaulay but not shellable)**.** Our exhaustive search found 3,247,490 graphs on 11 vertices where $\mathrm{Ind}(G)$ is Cohen-Macaulay over $\mathbb{Q}$. Because all but two of these graphs are also shellable, 3,247,488 of these graphs have the property that $\mathrm{Ind}(G)$ is Cohen-Macaulay over all fields $\mathbb{K}$. The two graphs where $\mathrm{Ind}(G)$ is Cohen-Macaulay but not shellable are the graphs G_1 and G_2 as given in Table 1. In this case, $\mathrm{Ind}(G_i)$ for $i = 1, 2$ is Cohen-Macaulay when $\mathrm{char}(\mathbb{K}) \neq 2$, but not shellable. The f-vector of $\mathrm{Ind}(G_1)$ and $\mathrm{Ind}(G_2)$ is $(1, 11, 30, 20)$.

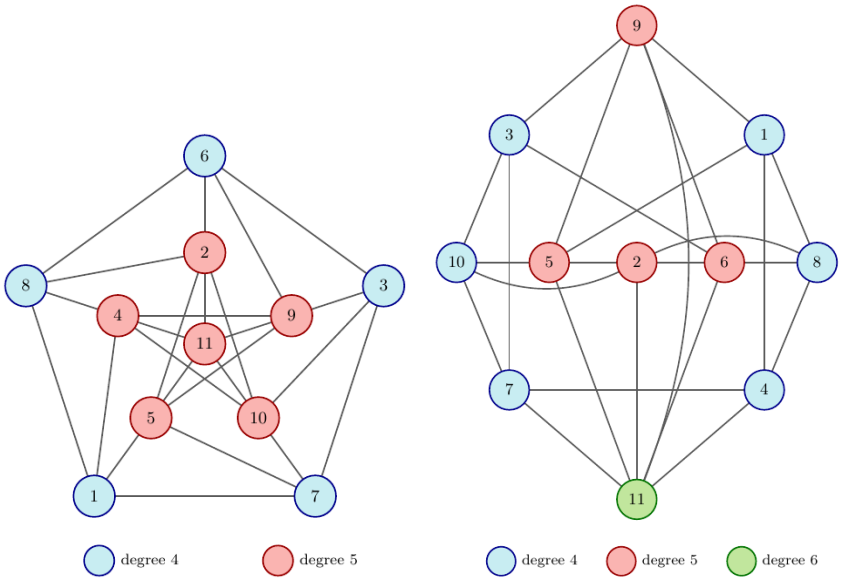


FIGURE 1. The two minimal graphs where $\mathrm{Ind}(G)$ is Cohen-Macaulay but not shellable. The graph G_1 is on the left and the graph G_2 is on the right.

Proof of Theorem 1.1 (2), (3). Recall that the reduced simplicial homology of $\mathrm{Ind}(G)$ only depends upon the characteristic of the field. Furthermore, by the universal coefficient theorem and Reisner's criterion, if $\mathrm{Ind}(G)$ is Cohen-Macaulay over some field $\mathbb{K}$, then $\mathrm{Ind}(G)$ is Cohen-Macaulay over $\mathbb{Q}$. Consequently, if G is a graph on 11 vertices where $\mathrm{Ind}(G)$ is Cohen-Macaulay over a field $\mathbb{K}$, then the graph G is among the 3,247,490 graphs found in our computer search which was done over $\mathbb{Q}$.

The independence complexes $\mathrm{Ind}(G_1)$ and $\mathrm{Ind}(G_2)$ both have the f-vector $(1, 11, 30, 20)$, so $\dim(\mathrm{Ind}(G_i)) = 2$ for $i = 1, 2$, and hence $\dim(R/I(G_i)) = 3$ for both graphs. Graph G_1 appears as the graph G_4 in Katzman's paper [13], while G_2 appears as G_3 in the same paper. Katzman showed that if $\mathrm{char}(\mathbb{K}) \neq 2$, then the projective dimension of $R/I(G_i)$ is 8, and is 9 if $\mathrm{char}(\mathbb{K}) = 2$ for $i = 1, 2$. By the Auslander-Buchsbaum Formula, the depth of $R/I(G_i)$ is then $11 - 8 = 3$ if $\mathrm{char}(\mathbb{K}) \neq 2$, and is 2 otherwise. So $\mathrm{Ind}(G_i)$ is Cohen-Macaulay if and only if $\mathrm{char}(\mathbb{K}) \neq 2$.

On the other hand, $\mathrm{Ind}(G_i)$ is not shellable for $i = 1, 2$. Indeed, if $\mathrm{Ind}(G_i)$ was shellable for either $i = 1, 2$, then it would be Cohen-Macaulay in all characteristics by Theorem 2.3. But we just showed this is not the case.

If G is any other graph on 11 vertices with $G \notin \{G_1, G_2\}$ and $\mathrm{Ind}(G)$ Cohen-Macaulay over some $\mathbb{K}$, and thus over $\mathbb{Q}$, then $\mathrm{Ind}(G)$ is also shellable. Indeed, if $\mathrm{Ind}(G)$ is not shellable, then G, G_1, G_2 would be three graphs whose independence complexes are Cohen-Macaulay over $\mathbb{Q}$ but not shellable. But our search only found two such graphs. So G must be among the 3,247,488 graphs whose independence complexes are shellable. Thus if $\mathrm{char}(\mathbb{K}) \neq 2$, G_1 and G_2 are the only graphs whose independence complexes are Cohen-Macaulay but not shellable. However, if $\mathrm{char}(\mathbb{K}) = 2$, an independence complex of a graph on 11 vertices is Cohen-Macaulay if and only if it is shellable. □

Remark 3.2. Both G_1 and G_2 have previously appeared in the literature, but it appears it went unnoticed that these graphs also provide minimal examples of graphs whose independence complexes are Cohen-Macaulay but not shellable.

As observed in the above proof, our graphs G_1 and G_2 appear as examples G_4 and G_3 in the appendix of Katzman's paper [13]. In his paper, Katzman showed that the edge ideals of G_3 and G_4 both have graded Betti numbers that can change depending upon the characteristic of the field $\mathbb{K}$.

The graph G_1 also appears in Jaramillo–Villarreal's paper [12, Example 5.4] on the $\mathtt{v}$-number of the edge ideal; this graph is used to give the first example of a graph whose edge ideal has a $\mathtt{v}$-number that is larger than its Castelnuovo-Mumford regularity, providing a counterexample to a conjecture of Núñez-Betancourt–Pitones–Villarreal [18]. This graph was further exploited by Saha–Van Tuyl [22, Theorem 3.1] to give a minimal example of a graph whose edge ideal has a $\mathtt{v}$-number greater than the degree of its h-polynomial.

The independence complexes of G_1 and G_2 are also related to the minimal clique complexes with torsion found by Adamaszek [1, Theorem 1]. More precisely, Adamaszek considers $\mathrm{Cl}(G)$, the clique complex of a graph, where the faces correspond to the cliques of the graph. It can be checked that $\mathrm{Cl}(G) = \mathrm{Ind}(\overline{G})$, where $\overline{G}$ denotes the complement of the graph G. His K_4 is the complement of our G_1, and his $K_2 = \overline{G}_2$.

As observed by Adamaszek, the complex $\mathrm{Cl}(\overline{G}_1)$ is homeomorphic to the two dimensional real projective plane $\mathbb{R}P^2$. Within the combinatorial commutative algebra community, the same example is sometimes referred to as *Terai's example*. Indeed, the complex $\mathrm{Ind}(G_1)$, which is attributed to Terai, appears as Exercise 6.3.65 in [27] as a triangulation of $\mathbb{R}P^2$ (or Exercise 5.3.31 of the 2001 edition of [27]).

Example 3.3 (Shellable but not vertex decomposable)**.** Among all the graphs on 11 vertices whose independence complexes are shellable, the independence complexes of the six graphs $H_1, \ldots, H_6$ are the only ones that fail to be vertex decomposable. The f-vector for $\mathrm{Ind}(H_i)$ is $(1, 11, 31, 21)$ for $i = 1, \ldots, 5$, while the f-vector of $\mathrm{Ind}(H_6)$ is $(1, 11, 30, 20)$. The graph H_6 is also the only example on 25 edges, while the other five graphs have 24 edges.

Proof of Theorem 1.1 (1). For each of the six graphs, one can use the `shellingOrder` command in the `SimplicialDecomposability` package [5] of *Macaulay2* to find a shelling order. As a concrete example, consider the graph H_1. The independence complex $\mathrm{Ind}(H_1)$ has shelling order

$$\{9,10,11\},\ \{1,10,11\},\ \{7,9,11\},\ \{2,7,11\},\ \{1,2,11\},\ \{8,9,10\},\ \{6,8,10\},$$
$$\{1,8,10\},\ \{7,8,9\},\ \{5,7,9\},\ \{4,5,9\},\ \{6,7,8\},\ \{1,3,8\},\ \{5,6,7\},\ \{2,6,7\},$$
$$\{4,5,6\},\ \{2,4,6\},\ \{1,2,3\},\ \{2,3,4\},\ \{3,4,5\},\ \{1,3,5\}.$$

On the other hand, if $\mathrm{Ind}(H_1)$ was vertex decomposable, then there would be a vertex $i \in V$ such that $\mathrm{del}_{\mathrm{Ind}(H_1)}(i)$ is vertex decomposable, and thus Cohen-Macaulay by Theorem 2.3. However, for $i = 1, 3, 5$, or 8, $\mathrm{del}_{\mathrm{Ind}(H_1)}(i)$ is not pure, and thus cannot be Cohen-Macaulay. For the vertices $2, 6$ and 9, the h-vector of $\mathrm{del}_{\mathrm{Ind}(H_1)}(i)$ is $(1, 7, 8, -1)$, for vertices $4, 10, 11$, the h-vector of the deletion is $(1, 7, 9, -1)$, and for $i = 7$ the h-vector is $(1, 7, 8, -2)$. So by Theorem 2.3 (3), this means none of the deletions are Cohen-Macaulay, and hence $\mathrm{Ind}(H_1)$ is not vertex decomposable.

Similar calculations show that $H_2, \ldots, H_6$ are shellable but not vertex decomposable. □

We can use the above examples to construct a graph on $n \geq 11$ vertices with similar properties. We recall the notion of a graph expansion. For a graph $G = (V, E)$ on n vertices, fix a tuple $\mathbf{s} = (s_1, \ldots, s_n) \in \mathbb{N}^n_{>0}$ of positive integers. The $\mathbf{s}$-*expansion of* G, denoted $G^{\mathbf{s}}$, is the graph on the vertex set $\{(i, j) \mid 1 \leq i \leq n,\ 1 \leq j \leq s_i\}$ with edge set

$$\{\{(i_1, j_1), (i_2, j_2)\} \mid \{i_1, i_2\} \in E \text{ or } i_1 = i_2\}.$$

That is, $G^{\mathbf{s}}$ is formed by replacing the vertex i with a clique of size s_i, and if $\{i, j\}$ is an edge in the original graph, we join all vertices in the two corresponding cliques.

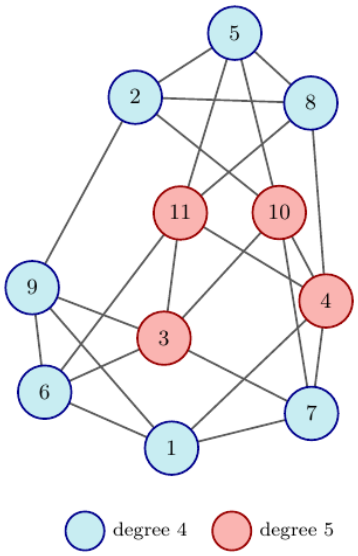
Theorem 3.4. *Let G be a graph and $\mathbf{s} \in \mathbb{N}^n_{>0}$.*

(1) [16, Theorem 2.7] $\mathrm{Ind}(G)$ *is vertex decomposable if and only if* $\mathrm{Ind}(G^{\mathbf{s}})$ *is vertex decomposable.*
(2) [16, Theorem 2.12] *If* $\mathrm{Ind}(G)$ *is shellable, then* $\mathrm{Ind}(G^{\mathbf{s}})$ *is shellable.*
(3) [4, Corollary 6.4] [21, Theorem 2.4] $\mathrm{Ind}(G)$ *is Cohen-Macaulay over* $\mathbb{K}$ *if and only if* $\mathrm{Ind}(G^{\mathbf{s}})$ *is Cohen-Macaulay over* $\mathbb{K}$.

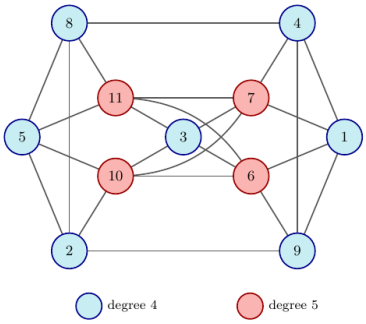
Remark 3.5. The authors of [16, 21] study the expansion of an arbitrary simplicial complex Δ, which is defined in terms of the facets of Δ. This definition of an expansion is equivalent to the inflation of a simplicial complex found in the paper [4]. We have specialized these definitions for $\mathrm{Ind}(G)$ so that the expansion/inflation can also be described in terms of the graph $G^{\mathbf{s}}$. Note that [16] conjectured that Theorem 3.4 (3) was true, but this result was already known [4].

Theorem 3.6. *For all integers $n \geq 11$ there exists a graph G on n vertices such that* $\mathrm{Ind}(G)$ *is Cohen-Macaulay but not shellable, respectively, shellable but not vertex decomposable.*

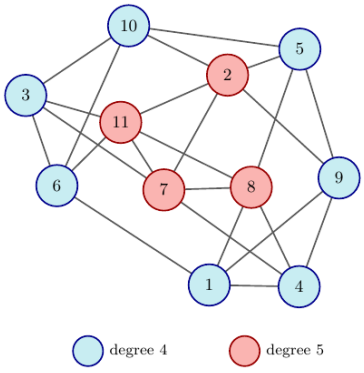
Proof. For a fixed integer $n \geq 11$, let $\mathbf{s} = (n - 10, 1, \ldots, 1) \in \mathbb{N}^{11}_{>0}$. Let G_1 be the graph of Example 3.1 and H_1 be the graph of Example 3.3. So, both $G_1^{\mathbf{s}}$ and $H_1^{\mathbf{s}}$ are graphs on n vertices. By Theorems 1.1 and 3.4, $\mathrm{Ind}(G_1^{\mathbf{s}})$ is Cohen-Macaulay over $\mathbb{K}$ if and only if $\mathrm{Ind}(G_1)$ is Cohen-Macaulay over $\mathbb{K}$ if and



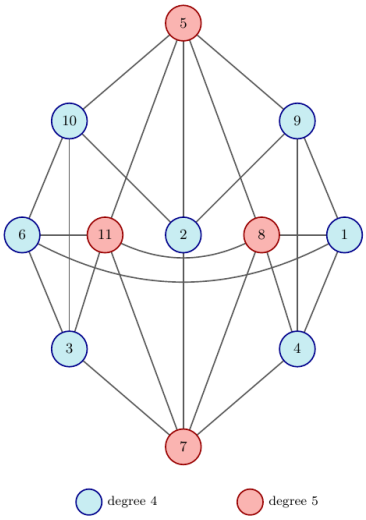
(A) Graph H_1



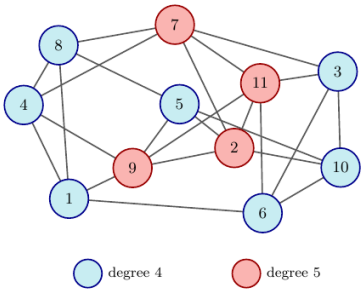
(B) Graph H_2



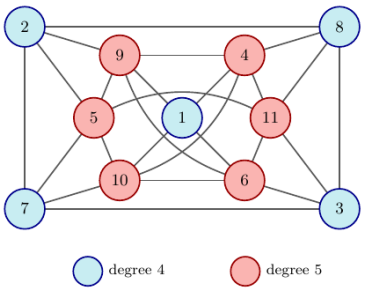
(C) Graph H_3



(D) Graph H_4



(E) Graph H_5



(F) Graph H_6

FIGURE 2. The six graphs G on 11 vertices with $\mathrm{Ind}(G)$ shellable but not vertex decomposable.

only if $\mathrm{char}(\mathbb{K}) \neq 2$. Consequently, $\mathrm{Ind}(G_1^{\mathbf{s}})$ is not shellable, because if it was, the complex would be Cohen-Macaulay over all fields.

Similarly, by Theorem 3.4, $\mathrm{Ind}(H_1^{\mathbf{s}})$ is shellable, but not vertex decomposable. □

4. Overview of the computer search

Table 2 summarizes our census of the connected graphs whose independence complexes belong to each of the following families: well-covered (pure), Cohen-Macaulay, shellable, and vertex decomposable. Note that the number of connected graphs and well-covered graphs on n vertices are given by the sequences A001349 [19] and A222625 [20], respectively. The first ten rows of the table were already computed by Baker–Vander Meulen–Van Tuyl [2]. Our contribution is the eleventh row.

n	Connected graphs	Well-covered	Cohen-Macaulay (over $\mathbb{Q}$)	Shellable	Vertex decomposable
1	1	1	1	1	1
2	1	1	1	1	1
3	2	1	1	1	1
4	6	3	2	2	2
5	21	6	5	5	5
6	112	27	20	20	20
7	853	108	82	82	82
8	11,117	788	565	565	565
9	261,080	9,035	5,688	5,688	5,688
10	11,716,571	196,928	102,039	102,039	102,039
11	1,006,700,565	7,797,877	3,247,490	3,247,488	3,247,482

Table 2. Number of connected graphs G on n vertices that are well-covered (i.e., $\mathrm{Ind}(G)$ is pure), and for which $\mathrm{Ind}(G)$ is Cohen-Macaulay over $\mathbb{Q}$, shellable, and vertex decomposable.

To ensure the accuracy of our results, we did our census over all connected graphs on 11 vertices twice. The first time, we used code written in *Macaulay2* [10] by the author, and the second time we used code written in Python with the help of Claude Opus 5.5. For the second run, we also computed the number of well-covered graphs in order to check our computations against the expected number given in A222625. In both runs, we checked whether a graph was Cohen-Macaulay over the field $\mathbb{Q}$.

The first computer search was carried out between July 2022 and June 2023 while the author was on sabbatical. The *Macaulay2* code made use of the packages `EdgeIdeals` [9], `Nauty` [6], and `SimplicialDecomposability` [5]. Note that the `Nauty` package is a *Macaulay2* package based upon the nauty program of McKay–Piperno [15]. The computations made use of the supercomputer clusters available to members of Digital Research Alliance of Canada (DRAC). The following work-flow was employed. Using the `Nauty` package, we generated all connected graphs on 11 vertices and e edges where $e \in \{10, \ldots, 55\}$. After further partitioning this set of graphs into files containing approximately 50,000 graphs, we first checked which of these graphs were Cohen-Macaulay. Those that were Cohen-Macaulay were then checked for vertex decomposability. A graph that failed to be vertex decomposable was then further checked whether it was shellable or not. While this search took advantage of some parallelization, setting up the relevant computations and keeping track of the output took up a significant amount of work time. All the examples of this paper were found during the first run in 2022–23.

The second search took place in September 2026, to validate that all the examples found in the first search formed a complete list of minimal examples. Claude Opus 5.5 was prompted to write code in Python (and the nauty program of McKay–Piperno) to repeat the census on 11 vertices. Additionally, the AI model was also prompted to take advantage of parallelization using the facilities of DRAC. The second run took a day to complete after it was recommended to exploit Theorem 2.3 (3) and make use of memoization when testing vertex decomposability. The run returned the exact same eight graphs described in this paper.

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Availability of Data. The list of all connected graphs on 11 vertices whose independence complexes are Cohen-Macaulay over $\mathbb{Q}$ is available at https://vantuyl.github.io/minimal-cm/.

Declaration of AI Use. The author made use of Claude Opus 5.5 to help generate the figures, tables, and improvements for the writing. As mentioned in Section 4, AI was also used to create code to cross-check the results of the first run.

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